

Review Problems

Dummit–Foote, Chapters 1 and 2

Review and Algebra Qualifier Preparation

1. Let x be an element of finite order n in a group G .

(a) Prove that

$$|x^k| = \frac{n}{\gcd(n, k)}.$$

(b) Prove that $\langle x^k \rangle = \langle x \rangle$ if and only if $\gcd(n, k) = 1$.

(c) Determine all generators of a cyclic group of order 30.

2. Let $G = \langle x \rangle$ be cyclic of order n .

(a) Prove directly that every subgroup of G is cyclic.

(b) Prove that for every positive divisor $d \mid n$, the group G has exactly one subgroup of order d .

3. Recall that a presentation of the dihedral group is given by

$$D_n = \langle r, s \mid r^n = s^2 = e, srs = r^{-1} \rangle.$$

(a) Determine $Z(D_n)$, distinguishing between odd and even n .

(b) Determine the centralizer $C_{D_n}(s)$.

4. Let $\sigma \in S_n$.

(a) Prove that

$$\sigma(a_1 a_2 \dots a_k) \sigma^{-1} = (\sigma(a_1) \sigma(a_2) \dots \sigma(a_k)).$$

(b) Let $\tau = (12 \dots n)$. Prove that

$$C_{S_n}(\tau) = \langle \tau \rangle.$$

(c) Deduce that the centralizer of an n -cycle in S_n has order n .

5. Recall that

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}, \quad i^2 = j^2 = k^2 = ijk = -1.$$

(a) Determine every subgroup of Q_8 .

(b) Prove directly that every subgroup of Q_8 is normal.

(c) Explain why this does not imply that Q_8 is abelian.

6. Let F be a field (feel free to take $F = \mathbf{Q}, \mathbf{R}$ or $\mathbf{C}$) and let

$$U = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in F \right\}.$$

- (a) Prove that U is a subgroup of $GL_3(F)$.
- (b) Derive explicit multiplication and inverse formulas in terms of the coordinates (a, b, c) .
- (c) Determine the center $Z(U)$.
- (d) Decide if the group U is abelian for any of the three fields $F = \mathbf{Q}, \mathbf{R}$ or $\mathbf{C}$.

7. Let $C_m = \langle x \rangle$ and $C_n = \langle y \rangle$.

- (a) Prove that a homomorphism $\varphi : C_m \rightarrow C_n$ is completely determined by $\varphi(x)$.
- (b) Determine precisely which elements of C_n can occur as $\varphi(x)$.
- (c) Prove that the number of homomorphisms $C_m \rightarrow C_n$ is $\gcd(m, n)$.
- (d) Determine when an injective or surjective homomorphism $C_m \rightarrow C_n$ exists.

8. Let G act on itself by left multiplication:

$$g \cdot x = gx.$$

- (a) Prove that this is a group action.
- (b) For each $g \in G$, let $L_g : G \rightarrow G$ be given by $L_g(x) = gx$. Prove that L_g is a permutation of G .
- (c) Prove that

$$G \longrightarrow S_G, \quad g \longmapsto L_g,$$

is an injective homomorphism.

- (d) Write the resulting permutation representation explicitly when $G = C_4$.

9. Let $C_m = \langle x \rangle$ and $C_n = \langle y \rangle$ be cyclic groups of order m and n respectively.

- (a) Prove that the order of $(x, y) \in C_m \times C_n$ is $\text{lcm}(m, n)$.
- (b) Prove that $C_m \times C_n$ is cyclic if and only if $\gcd(m, n) = 1$.
- (c) When $\gcd(m, n) = 1$, determine all generators of $C_m \times C_n$.

10. Let

$$r = (1\ 2\ \dots\ n), \quad s = (1\ 2) \in S_n, \quad n \geq 2.$$

- (a) Compute $r^k s r^{-k}$ for $0 \leq k \leq n-2$.
- (b) Prove that every adjacent transposition

$$(1\ 2), (2\ 3), \dots, (n-1\ n)$$

belongs to $\langle r, s \rangle$.

- (c) Prove that every permutation is a product of adjacent transpositions.
- (d) Conclude that

$$S_n = \langle (1\ 2\ \dots\ n), (1\ 2) \rangle.$$